

Process Specialist Evaluation

Section A: Multiple Choice Questions (25 points)

1. Which of the following parameters is most critical to monitor during the exponential growth phase of mammalian cell culture?
 - a) Lactate accumulation
 - b) Dissolved oxygen (DO)
 - c) Cell viability
 - d) All of the above
2. In a CHO cell culture process for monoclonal antibody production, which feeding strategy would most likely increase final titer?
 - a) Continuous feeding with constant glucose concentration
 - b) Temperature shift to 32°C during stationary phase
 - c) Maintaining high dissolved oxygen throughout culture
 - d) Increasing media osmolality during production phase
3. The most common first capture step in monoclonal antibody purification is:
 - a) Hydrophobic interaction chromatography
 - b) Ion exchange chromatography
 - c) Protein A affinity chromatography
 - d) Size exclusion chromatography
4. Which viral clearance method works primarily by size exclusion?
 - a) Low pH incubation
 - b) Nanofiltration
 - c) Solvent/detergent treatment
 - d) UV irradiation
5. Which of the following is NOT typically considered an upstream process?
 - a) Seed train expansion
 - b) Media optimization

- c) Cell line development
 - d) Tangential flow filtration
6. The primary purpose of a diafiltration step in downstream processing is to:
 - a) Remove host cell proteins
 - b) Exchange buffer/adjust formulation
 - c) Concentrate the product
 - d) Remove viral contaminants
 7. When scaling up a bioreactor from 5L to 500L, which parameter is most critical to maintain constant?
 - a) Agitation speed (RPM)
 - b) Power input per volume (P/V)
 - c) Volumetric gas flow (vvm)
 - d) Impeller diameter
 8. Which analytical technique is most commonly used to assess protein aggregation in a purified monoclonal antibody?
 - a) Reverse-phase HPLC
 - b) Size exclusion chromatography
 - c) Capillary electrophoresis
 - d) Mass spectrometry
 9. In biosimilar development, which of the following is the most important consideration during process development?
 - a) Maximizing yield
 - b) Minimizing cost
 - c) Matching critical quality attributes to the reference product
 - d) Minimizing process time
 10. What is the primary purpose of implementing Quality by Design (QbD) principles in bioprocess development?
 - a) To identify critical process parameters and their acceptable ranges
 - b) To reduce documentation requirements
 - c) To eliminate the need for in-process controls
 - d) To bypass validation requirements
 11. Which cell line is most commonly used for commercial production of monoclonal antibodies?
 - a) HEK293
 - b) CHO

- c) NS0
 - d) BHK
12. During process characterization, a Design of Experiments (DoE) approach is used to:
- a) Validate equipment performance
 - b) Determine process parameter ranges that ensure consistent product quality
 - c) Replace traditional validation methods
 - d) Comply with regulatory requirements only
13. The main advantage of single-use bioreactors compared to stainless steel bioreactors is:
- a) Higher cell density and productivity
 - b) Better oxygen transfer
 - c) Reduced cleaning and sterilization requirements
 - d) Higher maximum working volume
14. What is the primary objective of the viral inactivation step in monoclonal antibody purification?
- a) Improving product purity
 - b) Enhancing product stability
 - c) Ensuring product safety
 - d) Increasing process yield
15. In a fed-batch culture, what typically triggers the implementation of a temperature shift strategy?
- a) When cell density reaches its peak
 - b) When viability begins to decline
 - c) At the transition from growth to production phase
 - d) After a specific concentration of lactate is detected

Section B: Short Answer Questions (35 points)

1. Describe three key parameters you would optimize during process development of a biosimilar monoclonal antibody to ensure comparability with the reference product. (5 points)
2. Explain how you would troubleshoot a sudden decrease in cell viability during the production phase of a CHO cell culture. Include at least three potential causes and their corresponding solutions. (7 points)
3. Outline a typical downstream purification process for a monoclonal antibody, from harvest to final formulation, explaining the purpose of each step. (8 points)
4. Describe how you would approach process scale-up from a 10L bioreactor to a 500L production scale. What critical parameters would you maintain, and what challenges might you anticipate? (7 points)

5. Explain the concept of process characterization and how it contributes to establishing a control strategy for a bioprocess. (8 points)

Section C: Scenario-Based Questions (40 points)

1. Your team has been tasked with developing a biosimilar version of a commercially available monoclonal antibody. During initial development, you observe that your product has a different glycosylation profile compared to the reference product.
 - a) What upstream and downstream factors might contribute to this difference? (5 points)
 - b) Propose a systematic approach to modify your process to achieve a more similar glycosylation pattern. (5 points)
 - c) What analytical methods would you use to characterize and compare the glycosylation profiles? (5 points)
2. During routine manufacturing of a biosimilar product, you notice that the yield from Protein A chromatography has decreased by 25% compared to previous batches.
 - a) Describe a structured approach to investigating the root cause of this issue. (5 points)
 - b) List three potential causes for this decrease in yield and how you would confirm each hypothesis. (6 points)
 - c) Assuming you identify the root cause, outline the steps you would take to implement a corrective action and prevent recurrence. (4 points)
3. Your company is considering implementing continuous processing for a new biosimilar product.
 - a) Compare and contrast batch, fed-batch, and continuous upstream processing approaches. (5 points)
 - b) Describe two potential advantages and two potential challenges of implementing continuous downstream processing for monoclonal antibody purification. (5 points)